

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Credit System) Degree Examinations

January 2023

19ME8X08 – INDUSTRIAL POLLUTION CONTROL

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) What are the steps taken after 1952 great smog? Distinguish between carbon credit and carbon auditing.
- b) Calculate plume rise from 100m stack of 5m diameter when the flue gas exit velocity is 13.5 m/s and the flue gas temperature is 145° C. The atmospheric lapse rate is – 0.5°C/100m. It's a clear night with the wind speed of 4 m/s at stack mouth. Ambient temp is 15°C at stack height.
2. a) Explain the term global warming and ozone hole. Mention the impact of these on environment
- b) What is the function of atmosphere? Explain various layers of atmosphere with the help of a temperature variation plot
3. a) What is air pollution? Distinguish between
 - i) Natural and Man-made air pollutants
 - ii) Primary and Secondary air pollutants
 Give suitable examples for each.
- b) Estimate the maximum GLC of SO₂ at 1 km and 5 km downwind of power plant burning 9080 tons of coal per day containing 1% sulphur. Stack height is 250m, day is clear and sunny with wind speed of 3 m/s at 10m height. Hourly value of diffusion coefficient estimated by meteorological department are:

X	σ_y	σ_z
1km	140m	125m
5km	540m	500m

10 L3 2 2

Unit – II

4. a) Discuss about various sources of particulate matter.
- b) List the advantages and disadvantages of wet scrubber
- c) Explain various types of smoke. With neat sketch enumerate the working of smoke scope.
5. a) With a diagram explain the working principle of single stage ESP.
- b) What are the ill effects of oxides of nitrogen? Explain any three methods of controlling NO_x
6. a) With a neat sketch explain working of a cyclone separator. Mention the advantages and disadvantages of it.
- b) Explain how smoke rising from stacks be compared using Ringelmann Smoke Chart. Illustrate with help of the Ringelmann smoke table.

5 L2 3 1

5 L1 3 1

10 L2 4 1

10 L2 3 1

10 L2 4 1

10 L2 3 1

10 L2 4 1

P.T.O.

Unit – III

7.	a)	Discuss about various causes of water pollution.	5	L2	5	1
	b)	Enumerate about the problems associated with nuclear power plant with an example.	10	L2	5	1
	c)	List various sources of odour	5	L1	5	1
8.	a)	Explain working of a venture scrubber with a neat sketch. Discuss about the performance of it	10	L2	5	1
	b)	What do you mean by noise pollution? What are the ill effects of noise pollution? Explain various methods of controlling it	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
